ENEMY USE OF THE LARGEST NON-NUCLEAR MUNITION



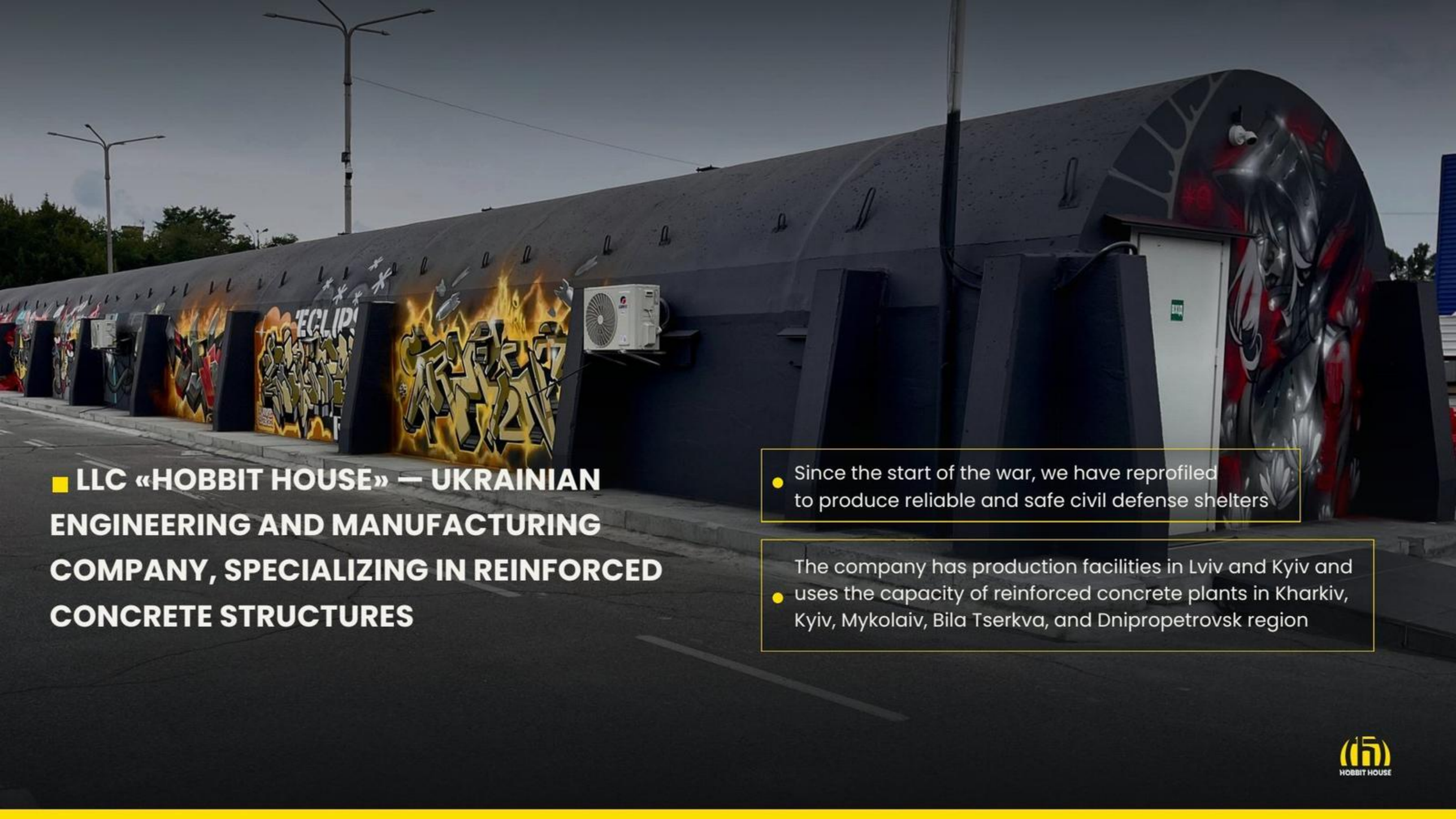
Vovchansk, 02.10.2024

CITY DESTROYED. STRUCTURE — INTACT



Vovchansk, 02.10.2024

HOBBIT HOUSE



LLC «HOBBIT HOUSE» — UKRAINIAN ENGINEERING AND MANUFACTURING COMPANY, SPECIALIZING IN REINFORCED CONCRETE STRUCTURES

- Since the start of the war, we have reprofiled to produce reliable and safe civil defense shelters

- The company has production facilities in Lviv and Kyiv and uses the capacity of reinforced concrete plants in Kharkiv, Kyiv, Mykolaiv, Bila Tserkva, and Dnipropetrovsk region

HOBBIT HOUSE

OUR SHELTERS



Underground modular shelters

TYPE «HOBBIT»



Above-ground modular civil defense protective structures

TYPE «FORTETSIA»

UNDERGROUND MODULAR SHELTERS TYPE «HOBBIT»

A modular underground shelter whose modules
can be assembled side by side or end to end









- Installed semi-recessed or fully underground.
- Complies with current standards DBN V.2.2-5:2023* and DSTU 9329:2025*.
- Used for construction of various types of underground civil defense shelters.
- Installed to protect employees and visitors of kindergartens, schools, healthcare facilities, etc.
- Hundreds of shelters are already installed across Ukraine, protecting people from shelling from Lviv to Kharkiv and Kherson.

*DSTU 9329:2025 — National Standard of Ukraine | *DBN V.2.2-5:2023 — State Building Norms of Ukraine

ABOVE-GROUND MODULAR PROTECTIVE STRUCTURE TYPE "FORTETSIA"

[Standard shelters are manufactured with a protection class for air blast load of 100 kPa]

500 kPa

IN FIELD TESTS, THE STANDARD «FORTRESS» STRUCTURE SEQUENTIALLY WITHSTOOD BLAST WAVES OF 100 KPA, 300 KPA, AND UP TO 500 KPA

- The structure protects against penetration of fragments weighing up to 0.4 kg and up to 50 mm in size, moving at speeds of up to 1,000 m/s.

- Installed in high-traffic locations — at transport stops and enterprises to shelter employees.

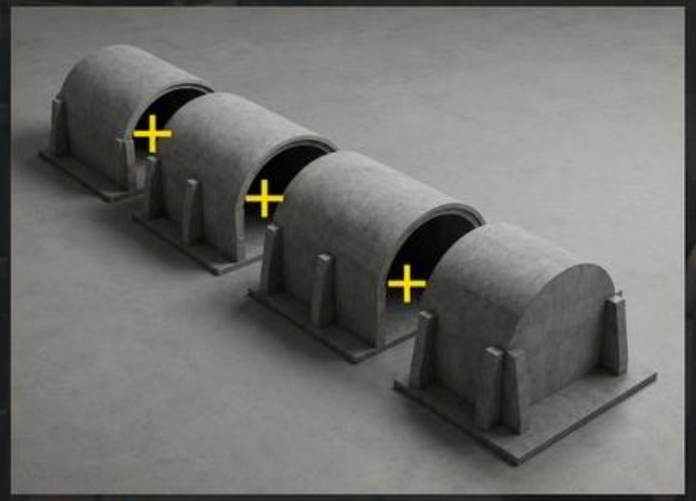




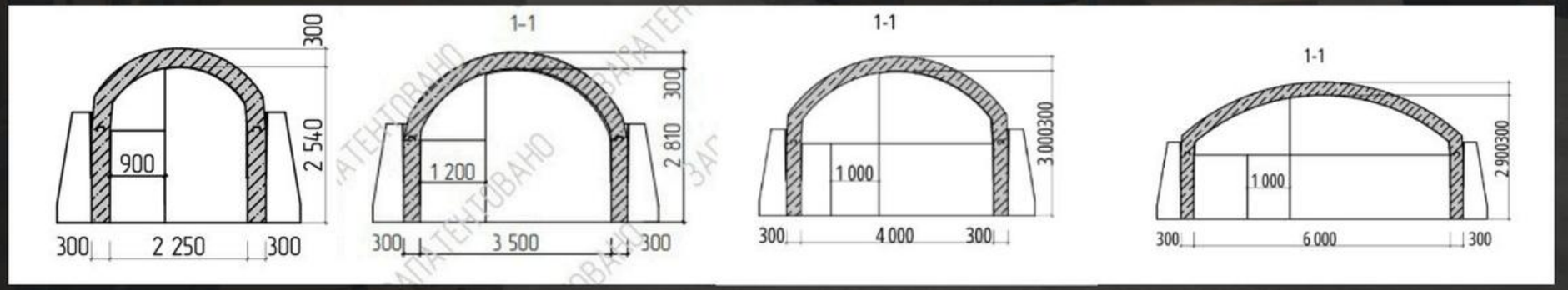

Examples of «Fortetsia» type shelter placement

"FORTETSIA" DUAL-USE

HOBBIT HOUSE OFFERS A DUAL-USE CONCEPT THAT TRANSFORMS A RELIABLE SHELTER INTO AN ACTIVE SPACE WORKING FOR BUSINESS AND COMMUNITY EVERY DAY.



Thanks to the modular system and interior widths from 2.25 to 6.0 meters, our structures are free of spatial constraints.



"FORTETSIA" DUAL-USE

For the commercial sector,
this means maximum efficiency:







DURING ALERTS, THE SHELTER CONTINUES TO OPERATE

spacious logistics hub • reliable warehouse • resilience hub • order pickup point • modern café

"FORTETSIA" DUAL-USE

At the same time, for municipalities this is ready-made social infrastructure where a multifunctional public space or educational center can be deployed.





In case of danger, the structure converts into a shelter in minimal time. And operations do not stop, as the key processes within it are already protected from external threats.

PRIMARY (MOBILE) SHELTERS

IN ACCORDANCE WITH DSTU 9329:2025:

LLC "Hobbit House" manufactures and installs primary (mobile) shelters (PMU) with internal module widths of:

- 2.25 meters
- 3.5 meters
- 4 meters
- 6 meters

FROM FORTRESS TYPE MODULES

SHELTER INSTALLATION VIDEO

Each shelter comes with a Certificate of Conformity DSTU 9329:2025 issued by the State Certification Center of the State Emergency Service of Ukraine, and a declaration in accordance with current legislation.



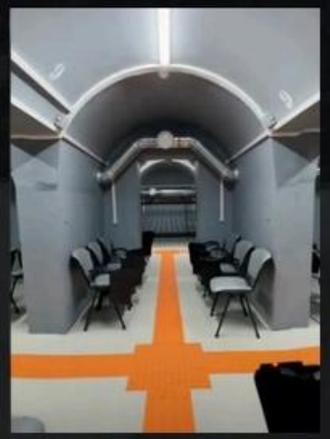


DESIGN AND CONSTRUCTION OF FALLOUT SHELTERS

Standard design and construction of fallout shelters from monolithic reinforced concrete takes up to a year, while modular structures from LLC «Hobbit House» significantly reduce construction time.

Just one week after work begins, the shelter can be ready for operation.









ADVANTAGES OF «HOBBIT HOUSE» MODULAR ARS STRUCTURES:

- Compliance with DBN V.2.2-5:2023 and DSTU 9195:2022
- Protection from penetrating radiation (Kz) 100, 200, 1000
- Use of arched roof, as the strongest form in construction;
- Fast installation by three to four workers instead of lengthy construction by multiple crews during standard reinforced concrete construction.

The speed of **above-ground** and **underground** ARS construction is demonstrated on video


FORTETSIA 2.25
FORTETSIA 3.5
FORTETSIA 4.0
FORTETSIA 6.0
Hobbit House

FORTETSIA 4.0 SHELTER

Our line of infrastructure solutions has been expanded with the «Fortetsia 4.0» shelter — an improved protective structure that has successfully passed full-scale tests and is fully certified under the new DSTU.





STRUCTURE INSTALLATION VIDEO

Unlike the basic models with widths of 2.25 and 3.5 meters, this generation has a width of

4 METERS

While keeping the overall length unchanged, we **significantly expanded its internal dimensions**, providing much more usable space for safe placement of people.

This increase in volume did not reduce strength: the object is **guaranteed to withstand all critical loads**, automatically resolving the issue of regulatory compliance.

This is a proven field-tested engineering protection tool that becomes the ideal platform before transitioning to even larger-scale solutions.

FORTETSIA 6.0 SHELTER

THE WIDEST IN THE SERIES IS THE «FORTETSIA 6.0» MODEL — THE FLAGSHIP SOLUTION FOR STRATEGIC-LEVEL FACILITIES.

Its key tactical advantage lies in **universality:** the object is equally effectively installed both deep underground and on the surface.

This scale is designed for the most complex autonomous tasks — from deploying protected barracks and command posts to logistics warehouses and production lines.

Strength and regulatory compliance are confirmed by official certificates of conformity from the State Certification Center of the State Emergency Service of Ukraine.

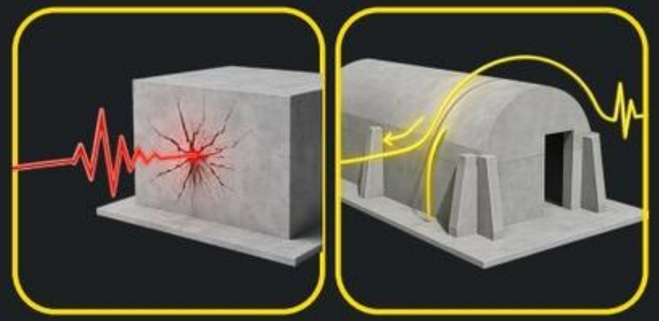
This model is used to create complex multi-layer shelters, providing protection against direct hits from enemy weapons.

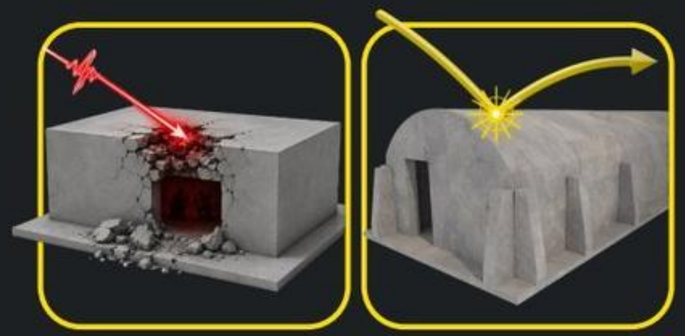


STRUCTURE INSTALLATION VIDEO

WHY HOBBIT HOUSE?

ARCHED ROOF





- Deflects the blast wave — doesn't absorb it, but redirects it
- An arch distributes load fundamentally better than a flat roof.
- Works like a ricochet: the roof shape forces the projectile to slide along the surface rather than penetrate it

BUTTRESSES



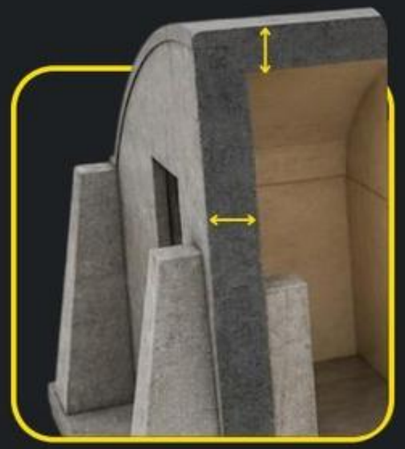
- Hold the structure under the blast wave and protect against the reverse pull of the vacuum wave
- During underground installation, they take on the lateral pressure and the load from the soil.
- Prevent deformation and tipping over

WHY HOBBIT HOUSE?



VERSATILITY

Above-ground, recessed,
or installation up to 6 meters
deep — a single structure



CUSTOMIZATION

Wall thickness increases
to meet individual customer
requirements





PROJECTS FOR PROTECTION OF OIL, GAS AND ENERGY INFRASTRUCTURE



BARBETTE

Prefabricated structure with buttresses. Used to protect tanks and containers.



BASTION

Prefabricated structure of concrete blocks, reinforced buttresses and arched block roof. Used to protect large-scale equipment.



CITADEL

Structure based on K-blocks with additional protection that can protect even against shaped-charge ammunition.

HTTPS://WWW.L7ENERGY.COM.UA/

FORTETSIA 4.0 AND 6.0

TEST 500 kPa

Shelter strength testing



FULL-SCALE TESTS FOR BLAST WAVE RESISTANCE

A series of 12 controlled detonations (6 per model) in three critical zones — **side wall, front side, roof center.**

Loads were increased gradually: **charge mass from 2 to 3 kg, distance from charge to sample — from 2.05 to 1.00 m**



REGULATORY BASIS:

DSTU 9329:2025. Standard level — **100 kPa**. Target test level — **500 kPa** (five times the standard).



RESULT:

Both model structures maintained integrity across all critical planes under loads from 100 to 500 kPa — **5 times above the standard.**

TEST VIDEOS

FORTETSIA 2.25 AND FORTETSIA 3.5

Tested at the testing ground



BLAST WAVE RESISTANCE

Another test was conducted by the State Research Institute for Testing and Certification of Weapons and Military Equipment and the State Emergency Service of Ukraine at a military testing ground.



REAL-WORLD TESTING

Shots were fired from:

- PG-7 (enhanced)
- IFV (30mm cannon)
- Tank T-72



TEST PASSED

The basic version of the Fortetsia, rated for a 100 kPa blast wave, successfully withstood consecutive strikes of 100 kPa, followed by 300 kPa, 500 kPa, and a 120mm mortar round. **The explosions caused no damage and the structure did not lose its full protective properties.**

VIDEO OF SUCCESSFULLY PASSED TEST

100 kPa • 300 kPa • 500 kPa

HOBBIT 2.0

152mm direct hit



TESTS:

- explosive charge (TNT 6 kg) placed in the ground near the wall;
- anti-tank mine TM-62 (TNT equiv. 7.6 kg) buried in the ground near the wall;
- explosive charge (TNT 6 kg) placed on the roof of the shelter.



RESULTS:

- no deformations leading to loss of protective properties;
- the structure retained its protective properties;
- no through-holes;
- the load-bearing capacity was not compromised and no structural destruction was observed;

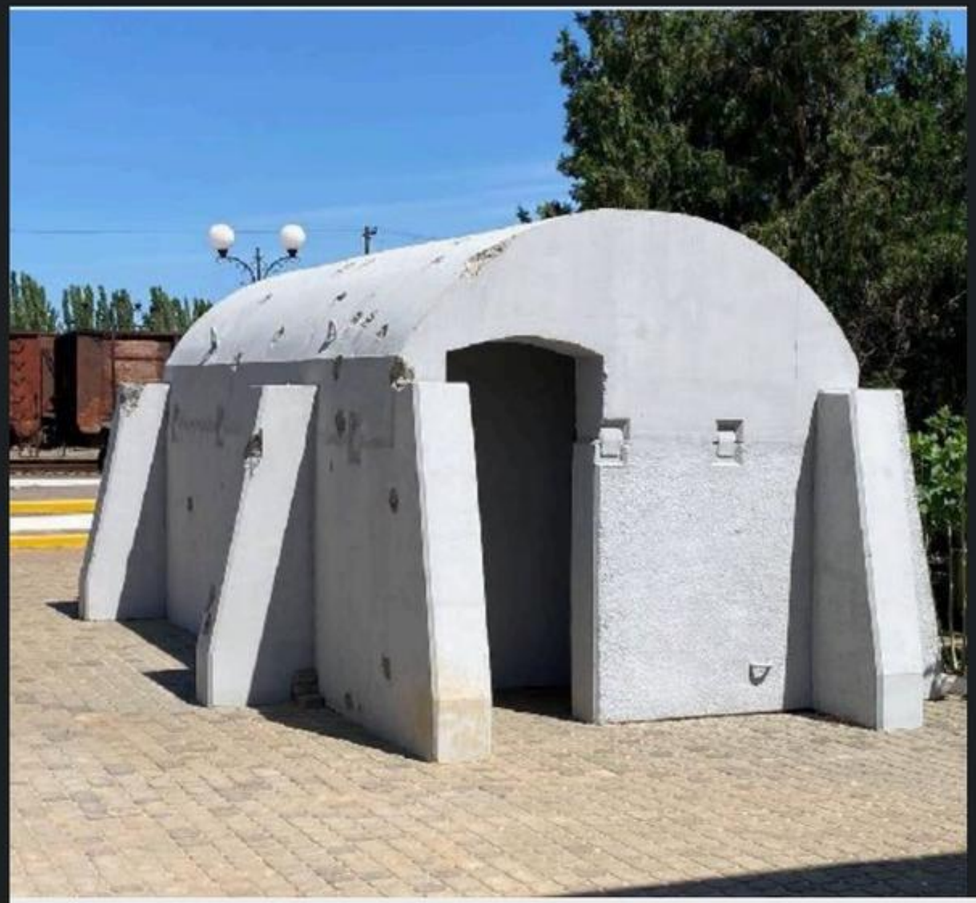
VIEW TESTS

REAL PROTECTION

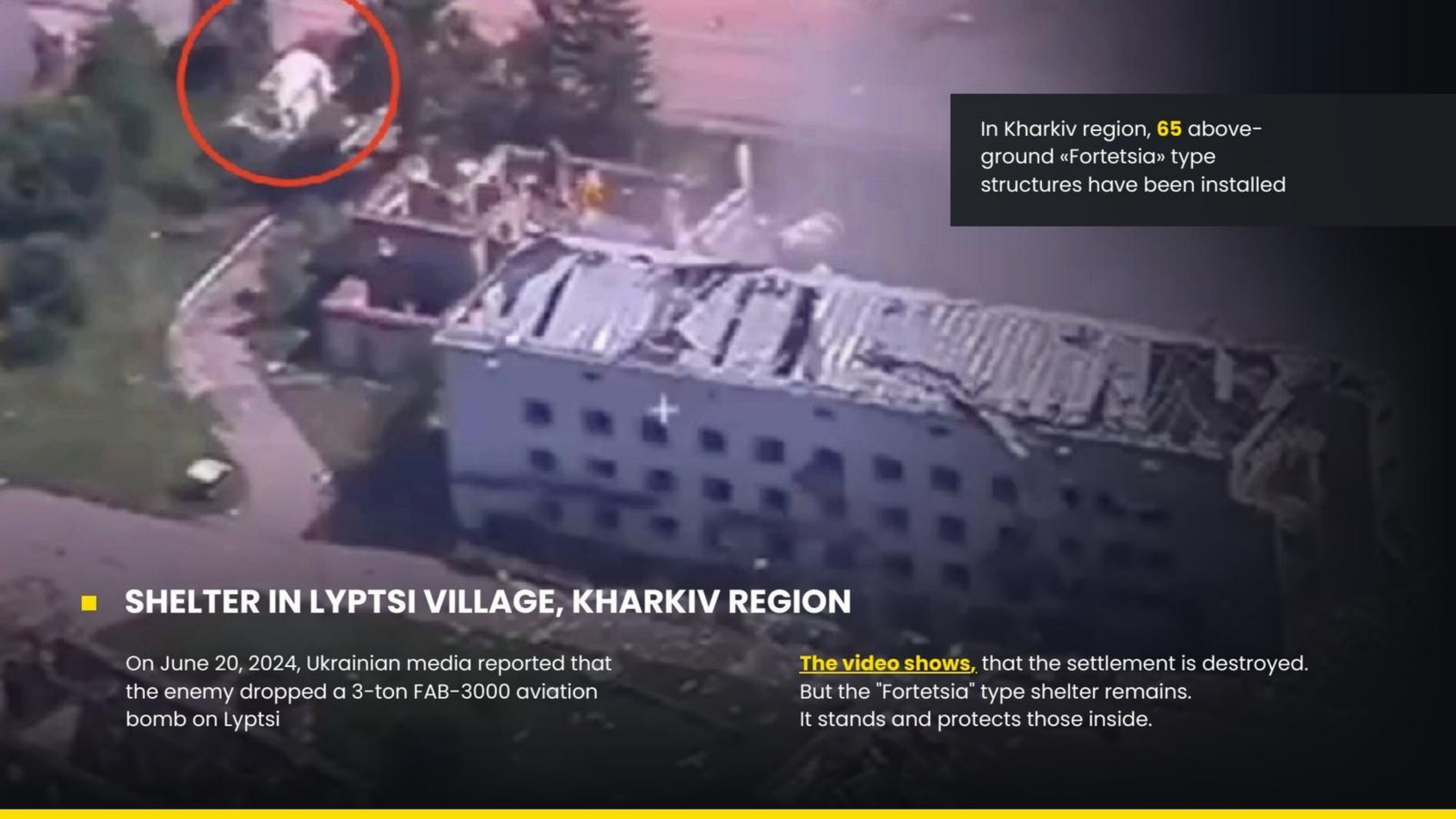
Above-ground modular civil defense shelters of the "Fortetsia" type by Hobbit House LLC are among the strongest manufactured in Ukraine. **This has been proven by:**

- calculations and conclusions of KNUBA — Ukraine's leading university in the field of construction;
- field tests **(link)**, during which "Fortetsia" type shelter withstood direct hits from T-72 tank high-explosive shells
- withstanding massive shelling at the hottest front-line locations (Kherson, Lyptsi, Vovchansk, etc.)

Result: maximum protection.
In the photo you can see one of 35 functioning shelters in Kherson, after the impact of a C-300 ballistic missile on the railway station. The explosion occurred 20 meters from the structure.



«Fortetsia» shelter at Kherson railway station
after a targeted enemy strike in May 2024



In Kharkiv region, **65** above-ground «Fortetsia» type structures have been installed

SHELTER IN LYPTSI VILLAGE, KHARKIV REGION

On June 20, 2024, Ukrainian media reported that the enemy dropped a 3-ton FAB-3000 aviation bomb on Lyptsi

The video shows, that the settlement is destroyed. But the "Fortetsia" type shelter remains. It stands and protects those inside.

REAL PROTECTION. SHELTER IN VOVCHANSK

One of the above-ground protective structures of the «Fortetsia» type is installed in Vovchansk, Kharkiv region

Above — photo of the city from 02.06.2024 from the MVS official page **showing the destroyed city**. The video shows the intact «Fortetsia» type shelter, still able to protect against shelling.



Photo below from 02.10.2024 — the same location in Vovchansk after the enemy dropped its largest non-nuclear munition on the city.



REAL PROTECTION. SHELTER IN VOVCHANSK



ON 02.10.24 THE MEDIA REPORTED:
the largest non-nuclear munition in the enemy's arsenal was dropped on Vovchansk.

The explosion **video shows** a terrifying blast wave spreading for kilometers from the epicenter, turning buildings and everything around to dust and ash.

Drone footage taken after the attack shows horrific destruction and piles of sand where buildings once stood. And the above-ground fortification civil defense protective structure of the «Fortetsia» type is intact.

CERTIFICATION AND EXPERIENCE

LLC «Hobbit House» has received a SIC (System of International Certification) certificate **confirming our compliance with NATO requirements** for design, development and production.

We also have experience implementing shelter installation projects **funded by USAID and UESF.** We acted as a subcontractor for the production and installation of shelters for DTEK and GRMU — the largest energy companies in Ukraine.

We have installed hundreds of shelters across Ukraine, from the front line to the western regions. Our shelters protect the defense industry, military personnel, employees of state and commercial enterprises, critical infrastructure workers, and civilians throughout Ukraine.


SYSTEM OF INTERNATIONAL CERTIFICATION

СЕРТИФІКАТ

«ХОББІТ ХАУС»

AQAP 2110:2016

«Вимоги НАТО щодо проектування, розробки та виробництва»


СЕРТИФІКАТ ВІДПОВІДНОСТІ

CERTIFICATE OF CONFORMITY

CONTACTS

HOBBIT HOUSE

WEBSITE hobbithouse.com.ua

YOUTUBE https://www.youtube.com/@hobbithouse

YOUR SAFETY IS OUR WORK